

MOUNIKA GUNTURU

Senior Data Engineer | Python | Cloud & Big Data

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SUMMARY

Senior Data Engineer with 8+ years of experience building scalable **ETL/ELT pipelines**, distributed data processing systems, and cloud-native data platforms for banking and financial institutions. Deep expertise in **Python, PySpark, Kafka, Airflow, Snowflake, and AWS** - processing TB-scale datasets and millions of daily transactions. **ISB-certified** in Business Analytics with a strong track record of reducing pipeline failure rates, **improving query performance by 40%**, and delivering near real-time insights for trading, risk, and regulatory reporting.

CERTIFICATIONS

Advanced Management Program in Business Analytics (AMPBA) - Indian School of Business (ISB), 2017–2018

EXPERIENCE

Senior Data Engineer | Oliver Wyman | New York, NY

08/2023 – Present

- Engineered **Python/FastAPI microservices** for risk modeling, scenario simulation, and portfolio optimization, integrating REST APIs with client trading systems.
- Built scalable ETL and real-time streaming pipelines using **PySpark, Pandas, Kafka, and Kinesis** - processing 10M+ daily transactions with sub-2s latency - integrated with Redshift and Snowflake.
- Designed **ETL/ELT** workflows with Apache Airflow and dbt to orchestrate financial data pipelines, improving reliability by 40% and reducing failure rates.
- Architected **AWS S3** Data Lake and Snowflake integrations to process and store 5TB+ daily financial datasets for downstream analytics.
- Built real-time ingestion pipelines **with Kafka and Spark** Structured Streaming for near real-time market risk monitoring across 50M+ daily events.
- Developed LLM-ready preprocessing pipelines for large financial datasets, supporting AI-driven predictive analytics and ML model integration.
- Optimized **PostgreSQL and MongoDB** databases, improving query performance by 40% for financial reporting across 100M+ record datasets.
- Containerized workloads with **Docker/Kubernetes** on **AWS (ECS, Lambda, RDS)**; automated infrastructure provisioning with **Terraform and Ansible**.
- Reduced computational overhead via **Celery/RabbitMQ** async task queues for high-volume transaction processing.
- Automated regulatory reporting workflows (SOX, Basel III), reducing manual audit prep time by 50%.
- Built **ReactJS + TypeScript** dashboards to visualize market risk and stress test results for executive decision-making.
- Implemented **CI/CD pipelines (Jenkins, GitHub Actions)** and real-time monitoring with **ELK Stack and Grafana**.
- Led a team of **4+ data engineers** to design and deliver scalable data pipelines, improving delivery efficiency and reducing pipeline failures by 40%.

Senior Data Engineer | Cummins | Columbus, IN

12/2021 – 07/2023

- Designed large-scale ETL pipelines **with Python, Kafka, Pandas, and SQLAlchemy** — processing 8M+ financial transactions daily - with PostgreSQL and Oracle query optimizations reducing report generation by 40%.
- Architected scalable data lake on AWS S3 with Snowflake integration for analytics on 3TB+ financial datasets.
- Orchestrated automated batch workflows with **Apache Airflow**, improving pipeline monitoring and reducing failure rates by 35%.
- Built dimensional data models (star and snowflake schemas) to support BI dashboards and advanced financial reporting.
- Introduced **Redis**-based caching for sub-second response times on analytics platforms handling 5M+ daily queries.
- Deployed scalable applications on **AWS (ECS, Lambda, RDS, S3)** with Terraform-automated infrastructure.
- Automated SOX/PCI-DSS regulatory reporting and audit workflows, reducing manual reporting efforts by 50%.
- Designed and implemented dimensional data models using **Kimball methodology**, including fact and dimension tables for financial analytics and reporting.
- Integrated ML models for predictive risk scoring, fraud detection, and asset performance forecasting.

- Built **ReactJS + TypeScript** dashboards to visualize portfolio performance, liquidity metrics, and financial KPIs.
- Streamlined **CI/CD via Jenkins** and **GitHub Actions**; conducted load/stress testing with **Locust** and **JMeter**.

Data Engineer | Banyan Data Services | Bangalore, India

08/2019 – 11/2021

- Orchestrated **Python microservices** on AWS with **Docker and Kubernetes**, ensuring **99.9% system** uptime across data pipeline infrastructure.
- Engineered distributed data processing workflows using **PySpark and Spark SQL**, transforming 2TB+ of structured and semi-structured enterprise datasets daily.
- Architected event-driven **Kafka** ingestion pipelines enabling near real-time analytics across downstream systems processing **30M+ events/day**.
- Established scalable data lake on **AWS S3 with Snowflake integration** for enterprise-scale analytics.
- Spearheaded migration from on-premise to cloud-native containerized platforms, reducing infrastructure costs by 25%.
- Implemented **RBAC and OAuth2** security models aligned with **HIPAA** and enterprise data compliance standards.
- Optimized data-heavy APIs with Redis caching and Celery task queues for near real-time processing.
- Automated CI/CD with **Jenkins** and **Terraform IaC**, cutting release cycles from weeks to days.

Data / Python Developer | Franklin Templeton | Hyderabad, India

03/2017 – 08/2019

- Built Python (Django/Flask) backend services and REST APIs for financial data ingestion, transformation, and real-time processing with 40% faster throughput.
- Developed **ETL pipelines** for portfolio analytics, trade reconciliation, and client reporting.
- Modernized legacy VBA/.NET tools into Python-based services, improving reliability and operational efficiency.
- Built ReactJS + TypeScript dashboards for fund managers with real-time investment performance insights.
- Ensured security with **OAuth2/JWT**, **Docker containerization**, and AWS deployment via Jenkins CI/CD.
- Aligned technology delivery with **SEBI and SOX** compliance standards in Agile sprints.

SKILLS

Languages & Frameworks: Python, SQL, TypeScript, Node.js, Django, Flask, FastAPI, ReactJS

Data Engineering: PySpark, Apache Spark, Kafka, Airflow, dbt, NiFi, CDC Pipelines, Batch & Streaming Processing, Data Modeling (Kimball), Data Lakehouse

Cloud & Storage: AWS (S3, EC2, RDS, Lambda, ECS, EKS), Azure (AKS, Blob), GCP, Snowflake, Redshift, Google BigQuery

Databases & Caching: PostgreSQL, Oracle, MySQL, MongoDB, Redis, CosmosDB, SQLAlchemy

Data Lake / Lakehouse: AWS S3 Data Lakes, Apache Iceberg, Delta Lake, Databricks

DevOps & CI/CD: Docker, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions, GitLab CI

Monitoring & Logging: ELK Stack (Elasticsearch, Logstash, Kibana), Grafana

Security & Compliance: OAuth2, JWT, RBAC, SOX, PCI-DSS, Basel III, HIPAA, SEBI, IAM

AI / ML Integration: LLM Data Pipelines, ML Model Integration, Predictive Analytics, Fraud Detection

Testing: PyTest, unittest, JMeter, Locust, API Testing

Frontend (supporting): ReactJS, TypeScript, D3.js, Plotly — used for internal dashboards and data visualization

BI & Visualization: Power BI / Tableau (basic dashboarding & reporting)

EDUCATION

Bachelor of Engineering, Electronics & Communication — Narayanamma Institute of Technology and Science, May 2015